

```

/*
ternary
alert(c ? t : f); // When c is true, there is no reason to evaluate f.
*/

```

```

var a = 4;
//--a;
var c = a*3; //9
console.log(c % 2);
var b = (a==5?'yes':'no'); //ternary
var d = Math.floor(Math.random()*1000);
console.log(d);

```

```

/*
JavaScript supports the following binary arithmetic operators:

```

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+   addition
-   subtraction
*   multiplication
/   division (returns a floating-point value)
%   modulus (returns the remainder)

```

```

JavaScript supports the following unary arithmetic operators:

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+   unary conversion of string to number
-   unary negation (reverses the sign)
++  increment (can be prefix or postfix)
--  decrement (can be prefix or postfix)

```

Assignment

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=   assign
+=  add and assign
-=  subtract and assign
*=  multiply and assign
/=  divide and assign
%=  modulus and assign

```

MATH

The Math object contains various math-related constants (for example, π) and functions (for example, cosine). (Note that the Math object has no constructor, unlike Array or Date. All its methods are "static", that is "class" methods.) All the trigonometric functions use angles expressed in radians, not degrees or grads.

Properties of the Math object

Property	Returned value	Description
Math.E	2.7183	e: Natural logarithm base
Math.LN2	0.69315	Natural logarithm of 2
Math.LN10	2.3026	Natural logarithm of 10
Math.LOG2E	1.4427	Logarithm to the base 2 of e
Math.LOG10E	0.43429	Logarithm to the base 10 of e
Math.PI	3.14159	π : circumference/diameter of a circle
Math.SQRT1_2	0.70711	Square root of $\frac{1}{2}$
Math.SQRT2	1.4142	Square root of 2

Methods of the Math object

Example Returned value

rounded to 5 digits Description

Math.abs(-2.3) 2.3 Absolute value: $(x < 0) ? -x : x$ Math.acos(Math.SQRT1_2) 0.78540 rad. = 45° ArccosineMath.asin(Math.SQRT1_2) 0.78540 rad. = 45° ArcsineMath.atan(1) 0.78540 rad. = 45° Half circle arctangent ($-\pi/2$ to $+\pi/2$)Math.atan2(-3.7, -3.7) -2.3562 rad. = -135° Whole circle arctangent ($-\pi$ to $+\pi$)Math.ceil(1.1) 2 Ceiling: round up to smallest integer $\geq$ argument

Math.cos(Math.PI/4) 0.70711 Cosine

Math.exp(1) 2.7183 Exponential function: e raised to this power

Math.floor(1.9) 1 Floor: round down to largest integer $\leq$ argument

Math.log(Math.E) 1 Natural logarithm, base e

Math.max(1, -2) 1 Maximum: $(x > y) ? x : y$ Math.min(1, -2) -2 Minimum: $(x < y) ? x : y$ Math.pow(-3, 2) 9 Exponentiation (raised to the power of): Math.pow(x, y) gives x^y

Math.random() 0.17068 Pseudorandom number between 0 (inclusive) and 1 (exclusive)

Math.round(1.5) 2 Round to the nearest integer; half fractions are rounded up (e.g. 1.5 rounds to 2)

Math.sin(Math.PI/4) 0.70711 Sine

Math.sqrt(49) 7 Square root

Math.tan(Math.PI/4) 1 Tangent

*/